

OPERATOR'S MANUAL

GDS Server

Global Discovery Server for OPC UA — where applications register themselves, get approved, and receive certificates from one built-in Certificate Authority.

Document GDS-OP-1 **Revision** 1.0 **Applies to** Python GDS Server, OPC UA Part 12 workflow

AT A GLANCE

Web admin UI	<code>https://<gds-host>:8443/</code> – ask your administrator for the exact address
OPC UA endpoint	<code>opc.tcp://<gds-host>:4840/gds/</code> – this is what applications connect to, not people
Default certificate group	<code>DefaultApplicationGroup</code>
Default certificate lifetime	2 years, unless your administrator changed it
Sign-in	see §3 – every installation starts with one admin account

1 What this server does

Before an OPC UA application can be trusted by anything else on the network, it needs an identity — an application instance certificate signed by a Certificate Authority everyone agrees to trust. This server is that authority, plus the paperwork trail around it.

Every certificate this server hands out follows the same four steps:



2 Two ways in

The server answers two completely different kinds of visitors on two different ports.

Web admin UI

For the person running the server: approve applications, inspect and revoke certificates, manage the trust list.

```
https://<gds-host>:8443/
```

OPC UA endpoint

For OPC UA client and server applications: registration and certificate requests, over the OPC UA binary protocol.

```
opc.tcp://<gds-host>:4840/gds/
```

You will spend almost all of your time in the web UI — the OPC UA endpoint is where the machines talk to each other.

3 Signing in

The first time the server starts, it creates one administrator account. If nobody set a password ahead of time, it generates one and prints it — once — to the server's startup log. Whoever installed the server should have that password; there's currently no self-service "forgot password" link, so if it's lost, an administrator has to reset it from the server itself.

Once signed in, you stay signed in until you log out or the server restarts (a persistent session key keeps you logged in across restarts if your administrator configured one).

4 The dashboard

The dashboard is the first thing you see after signing in — four counters and a running log:

- **Pending applications** — waiting on you, see §5.
- **Approved applications** — cleared to request certificates.
- **Active certificates** — currently valid and unrevoked.
- **Revoked certificates** — invalidated, and listed in the next trust list handed out.

Below that, **Recent activity** lists every registration, approval, issuance, and revocation in order — useful as a first stop when something doesn't look right.

5 Reviewing applications

This is the job you'll do most often. When an application registers itself, it appears on the **Applications** page carrying one of three states:

pending just registered, waiting on you **approved** may request and renew certificates
rejected blocked, but still on record

Open an application to see what it told the server about itself — its `ApplicationUri`, type (server, client, or both), and discovery URLs — then **Approve** or **Reject**. An application cannot get, renew, or check on a certificate until it's approved; every certificate-related request it makes is refused until then.

Unregister is different from rejecting: it erases the application's record entirely, including its certificate history. Use it for applications that are gone for good, not ones you're merely saying no to for now.

6 Certificates

Once approved, an application can ask this server for a certificate in one of two ways: bringing its own key pair and a signing request, or asking the server to generate the key pair too. Either way, the result lands on the **Certificates** page:

active currently valid **revoked** invalidated by an administrator

From there you can **download** a certificate (for inspection or for handing to another system) or **revoke** it. Revoking takes effect the moment another application next pulls the trust list — it doesn't reach into a connection that's already open elsewhere, so a compromised certificate should be revoked as soon as it's noticed.

7 Trust list & certificate authority

Every certificate this server issues is signed by one root CA, generated the first time the server ever started. Any other system that needs to trust GDS-issued applications only needs to trust that one root certificate — download it from the **Trust List / CA** page and install it wherever that trust needs to live.

The same page lets you:

- view the current revocation list (CRL), rebuilt automatically from whatever's marked revoked;
- upload additional trusted or issuer certificates — for example, another site's CA you want this server's applications to trust as well.

All of it — the root CA, the CRL, and anything uploaded — goes out together as one bundle whenever an application asks for the trust list.

8 Connecting an application

If you're bringing a new OPC UA client or server onto the network:

1. Point it at the OPC UA endpoint above and have it register itself — this happens automatically the first time most OPC UA stacks talk to a GDS.
2. It will sit **pending** until an administrator approves it (§5).
3. Once approved, it can request its certificate the way its own configuration expects — the server supports both a bring-your-own-request flow and a server-generated-key-pair flow.

The exact method names and parameters for integrators writing that registration logic — `RegisterApplication`, `StartSigningRequest`, `StartNewKeyPairRequest`, and so on — are documented in the project's technical README, not repeated here.

9 Troubleshooting

An application can't get a certificate.

Check the Applications page — it's almost always still pending. Approve it and have the application try again.

I forgot the admin password.

There's no self-service reset yet. Whoever administers the server itself will need to reset it there.

A device that used to connect fine now fails to authenticate.

Check whether its certificate expired or was revoked, under Certificates — the status column shows both.

Where's the deeper technical detail?

Ports, environment variables, and the security design behind all of this are covered in the project's README, aimed at the people installing and maintaining the server.

This manual covers day-to-day operation of the web admin UI. Installation, configuration, and integration details live in the project's technical documentation.